higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PRINCIPLES OF COMPUTER PROGRAMMING
(First Paper)
NQF LEVEL 3

4 MARCH 2019

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	D		
	1.1.2	C		
	1.1.3	B		
	1.1.4	C		
	1.1.5	D		
	1.1.6	C		
	1.1.7	B		
	1.1.8	C		
	1.1.9	D		
	1.1.10	A		
			(10 × 1)	(10)
1.2	1.2.1	True		
	1.2.2	True		
	1.2.3	False		
	1.2.4	False		
	1.2.5	False		
	1.2.6	False		
	1.2.7	False		
	1.2.8	True		
	1.2.9	False		
	1.2.10	True		
			(10 × 1)	(10)
1.3	1.3.1	Hexadecimal		
	1.3.2	Index out-of-range error		
	1.3.3	Structured Query Language (SQL)		
	1.3.4	One-to-one		
	1.3.5	Attribute/Property		
			(5 × 1)	(5)
				[25]

TOTAL SECTION A: 25

SECTION B

QUESTION 2

2.1 Step 1: Rewrite the original binary number
11100111_2

Step2: Change the zero's to be one's and the ones to be zero's
00011000_2 ✓

Step 3: Add 1 to step 2
00011000_2
$+\underline{00000001_2}$ ✓
$\underline{00011001_2}$ ✓✓ (4)

2.2 Y = NOT✓ (A AND✓ B OR✓ NOT✓ C) (4)

2.3 •A function returns a value
•A procedure returns no value. (2)

2.4 •A variable is memory location with a value that can be changed by the program.
•A constant is a memory location with a value that cannot be changed by the program. (2 × 2) (4)
[14]

QUESTION 3

3.1 **NB Answers must correspond to line numbers, as this is what is being tested.**

Line

–	Mark(i)	Mark() >89	Mark() >79	Mark() >69	Mark() >59	Mark() >49	Mark() >39	Other	Symbol	Output	
5	0✓										
6		20									
7			FALSE								
10				FALSE							
13					FALSE						
15						FALSE					
17							FALSE				
20								FALSE			
22									TRUE✓		
6										G	20 G✓
5	1✓										
6		90									
7			TRUE✓								
6										A	90 A✓
5	2✓										
6		45									
7			FALSE								
10				FALSE							
13					FALSE						
15						FALSE					
17							FALSE				
20								TRUE✓			
21										F	
6											45 F✓

(9 × 1) (9)

3.2
- Wrong variable type
- Incorrect field declaration
- Mixing variables types (Any 2 × 1) (2)

3.3 This refers to how many bits of data the computer handles at a time (2)

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[13]

QUESTION 4

4.1 4.1.1 SELECT *✓ FROM OurInfo✓
WHERE gender = M✓AND✓ Province = Gauteng✓ (5)

4.1.2. ID✓
It is unique/It is computer generated/No chance of duplication (2)

4.2 •A primary key is a key in a table that uniquely identifies a record.
•Alternate keys are unique keys in the table other than the primary key.
•A foreign key is a key that references the primary key of another table.
(3 × 2) (6)
[13]

QUESTION 5

5.1 1 × 2 = 2
2 × 2 = 4
3 × 2 = 6
4 × 2 = 8
5 × 2 = 10 (5)

5.2 •Sequence structure
•Iteration structure
•Decision structure (3)

5.3 •Input
•Processing
•Output (3)
[11]

QUESTION 6

6.1 *Defensive programming* is when the programmer anticipates an error✓ on the side of the user, puts a mechanism to handle that error in the code✓ and inform the user of the error.✓ (3)

6.2 6.2.1 These users need to find help with ease, so as a programmer make sure help is included in you program.

6.2.2 Disable the buttons and other controls when the program is busy.

6.2.3 Your user interface must allow a user to undo operations when possible.

6.2.4 Use status messages and easy access to help facilities.

6.2.5 Use shortcut keys and offer help when it is needed.
(5 × 2) (10)
[13]

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QUESTION 7

7.1 7.1.1
- False
- True (Any 1 x 1) (1)

7.1.2 +1.26585 × 10^{11} (Sign 1 + Decimal point 1) (2)

7.2 7.2.1
- It consists of a string of characters, making up words and numbers.
- It is sequential.
- It stores data in an unstructured or free format. (Any 2 × 1)

7.2.2
- It stores several values using a single variable name.
- It is temporary storage.
- It can only store data of one simple data type. (Any 2 × 1)

7.2.3
- Data is stored in fields and records.
- Data is stored in a set of tables.
- Each table has different keys.
- Tables can be joined in a relationship.
- It is usually supplied as part of a package. (Any 2 × 1)

(3 × 2) (6)

7.3 A general name for data types that don't fit any of the simple data types. (2)

[11]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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